

COSC 39: Theory of Computation

Problem P.rove that all following languages are undecidable.

1. Prove that there is no Turing machine that decides the language

$$\text{EQUAL} := \{(\langle M_1 \rangle, \langle M_2 \rangle) \mid L(M_1) = L(M_2) \text{ for Turing machine } M_1 \text{ and } M_2\}$$

2. Prove that there is no Turing machine that decides the language

$$\text{REG} := \{\langle M \rangle \mid L(M) \text{ is regular}\}$$